

1. It does point to the right lines of code. It also gives information about the memory address of the conflict and if there are locks held by the threads.
2. When one of the conflicting accesses is removed, the race condition disappears because there is no longer concurrent access to shared data. Helgrind no longer reports a race.
3. With only a lock on one of the lines of code, helgrind reports a possible data race condition, just like without any locks.
4. With locks around both, helgrind reports no possible data race conditions.
5. A deadlock is when threads are stuck waiting for their turn to use the resources and will never get them. This code has a deadlock because the two threads get the locks in a different order, thread 1 goes 1 then 2, while thread 2 goes 2 then 1. This makes it so both threads successfully get the first lock, but are then stuck waiting for their second one.
6. Helgrind reports that the lock order is violated after it has been set.
7. It does not have the same problem as the previous one because there is now a new lock that would make the second thread wait until it is unlocked before it can start locking. Helgrind should not be reporting the same error because this new lock is solving the problem. This tells us that tools like helgrind just look at the code for potential problems, not what would actually happen when it is run.
8. The parent ends up just taking up resources, spinning until the child says that it is done.
9. Helgrind reports a possible data race. The code is not correct because there is a potential data race caused when the child changes the value to 1 and the parent is checking at the same time.
10. This code is preferred because it now removes the possible race condition by locking the variable when it is either being changed by the child or checked by the parent. It is also more performant since the conditional variable makes the thread sleep when it is waiting, so it uses fewer system resources.
11. Helgrind does not report any errors on this version.